Autonomous IR Surface Anomaly Detection and Optimisation

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***Abstract*—Low cost infrared (IR) reflectance sensors offer a practical basis for autonomous surface inspection for applications such as quality control. However, their diffuse spatial sensing footprint limits their sensitivity to small or narrow surface defects. This research investigates whether a two-pass scanning strategy with speed adaptation and signal filtering can improve surface anomaly detection using the Pololu 3Pi+ robot. Experiments were conducted on printed maps with prescribed imperfection widths (1-5cm) and contrast levels (20-100%). Reducing the robot's passing speed from PWM=60 to PWM=20 increased boundary sampling density from approximately 1 to 6 samples. This enabled reliable detection down to 2cm. EMA signal filtering provided the best balance of noise reduction and real time performance. Whilst it improved noise characteristics, it could not recover spatial detail lost at higher speeds. Results confirm that motion control is the primary determinant of detection performance, with sensor geometry representing the fundamental minimum detectable size limit.**

I. Introduction

Autonomous robotic inspection has become increasingly common in industrial quality control and maintenance [1]. Mobile robots offer a flexible, cost-effective solution for identifying surface defects by detecting variations in surface reflectance via onboard sensors.

The Pololu 3Pi+ robot carries five infrared (IR) reflectance sensors that emit IR radiation and measure the reflected intensity via photodiodes. Surfaces with lower reflectivity produce reduced sensor responses. However, IR illumination is inherently diffuse and each sensor integrates reflectance over a spatial region rather than a point, limiting sensitivity to narrow defects. Noise and bias further require calibration and filtering for reliable measurements [2]. Detection performance is also coupled to robot motion. Traversal speed and sampling frequency directly influence how surface features are observed, creating a trade-off between spatial resolution and scan duration that signal processing alone cannot resolve.

This project investigates the feasibility of IR sensors for detecting small surface imperfections using the 3Pi+. A two-phase approach is used, consisting of a single-pass baseline scan followed by a structured two-pass strategy where, upon detecting an anomaly, the robot rescans at reduced speed to increase spatial sampling density. Specially designed maps simulate cracks of controlled width and contrast, enabling systematic characterisation of sensor limits.

The objective of this study is therefore to characterise the sensitivity and limitations of low cost IR reflectance sensors for robotic surface inspection. Specifically, the work seeks to determine the smallest detectable defect size, evaluate the robustness of detection under variations in robot trajectory and speed, and assess how sensor signal processing can improve detection reliability. By systematically analysing these factors, this study provides an insight into the suitability of low-cost mobile robotic platforms for autonomous inspection tasks

A. Hypothesis Statement

Having identified one of the weaknesses of the 3Pi+ IR sensors being that IR illumination is measured over a spatial footprint rather than a single point, we hypothesise that reducing the robot's passing speed when a surface anomaly is first detected will improve crack/imperfection size detection accuracy. Furthermore applying different low-pass signal filtering to the sensor will reduce noise detection and improve crack visibility.

II. Implementation

The codes and experimental maps used to carry out the experiments in this study can be found in the following GitHub Repository. The experiments have been carried out under controlled scenarios using maps with predefined surface imperfections as shown in figure 1. The system has been implemented on the 3Pi+ Robot to autonomously detect and characterise surface anomalies via its IR sensors. As the robot traverses a surface, variation in reflected IR intensity are used to identify anomalies and estimate both their spatial width and relative "contrast %".

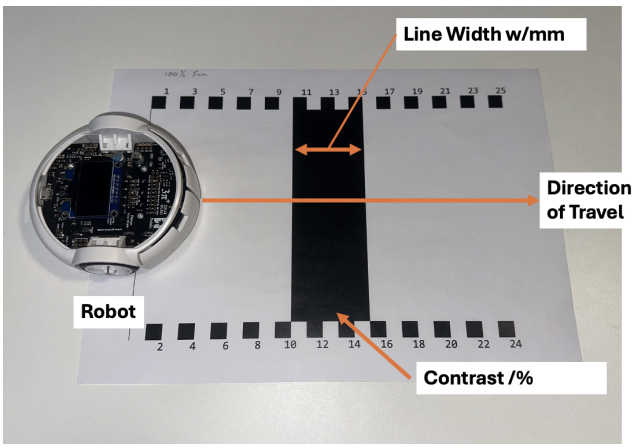

Fig. 1. 2D localisation - Experimental setup of robot on baseline map with imperfection width of 5cm and 100% Contrast

A. Initial Phase

The initial implementation, shown in Algorithm 1, focused on controlled motion and sensor data collection, providing the foundation for later improvements in anomaly classification and width estimation. The robot was driven forward at a fixed target distance of 200mm, chosen to ensure the robot traversed the full imperfection whilst

remaining within the map boundaries. IR sensor readings were continuously sampled and stored alongside encoder-estimated position, producing a positional sensor array for offline analysis (shown in Figure 2). Light (low contrast) values produce higher reading than dark (high contrast) values. A baseline 100% contrast 'Black' value was initially taken from the normal surface to be used as the reference for anomaly detection.

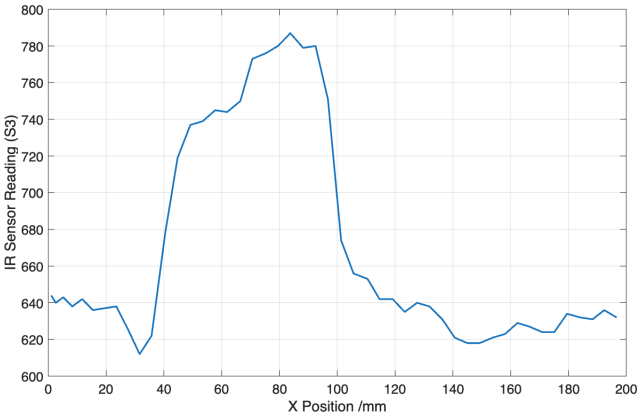

Fig. 2. Robot distance travelled against unfiltered IR sensor reading where the peak represents the highest anomaly contrast detected

The IR sensor readings are normalised against this baseline to reduce the effect of sensor bias and environmental variation, ensuring that changes in measured intensity correspond to genuine surface features rather than sensor inconsistency or ambient lighting effects. Whilst this enabled detailed offline analysis of sensor behaviour across the full range of test surfaces, onboard memory constraints limited the achievable sampling rate and total scan duration. This created a trade-off between spatial resolution and data length. Furthermore, this implementation did not support real-time anomaly detection or direct width estimation during traversal, and raw sensor readings were affected by noise which limited the reliability of simple threshold-based detection. These limitations directly motivated the development of the improved two-pass implementation.

Algorithm 1 Initial IR Surface Data Collection

Input: Raw IR sensor readings **Output:** Position-sensor array, R
Initialise motors, encoders, PID, and IR sensor
$targetX \leftarrow 200$, $index \leftarrow 0$
while robot has not reached $targetX$ **do**
    Update position; apply PID; drive forward
    $R[index] \leftarrow [x,$ read IR$]$, where x is current position
    $index \leftarrow index + 1$
end while
Stop motors
while true **do**
    Print "X,Raw"
    for $i \leftarrow 0$ to $index - 1$ **do**
        Print $R[i][0], R[i][1]$
    end for
    Delay 1000 ms
end while

Algorithm 2 Two-Pass IR Surface Anomaly Detection with Backtracking

Input: Filtered IR readings **Output:** Anomaly positions and widths
Initialise system (motors, encoders, PID, sensors)
First Pass
$scanArmed \leftarrow false$
while not armed or within detection window **do**
    Update position; apply PID; read IR
    if $reading < threshold$ **then**
        $scanArmed \leftarrow true$; record entry; update time
    end if
end while
Stop motors
Backtrack
Reverse to pre-detection position
Second Pass
$scanArmed \leftarrow false$
while not armed or within window **do**
    Update position; read IR
    if $reading < adaptive\ threshold$ **then**
        record entry
    else
        record exit
    end if
end while
Compute width; stop motors

B. Testing Phase

An improved implementation built on the initial data acquisition system by adding real time surface anomaly detection and measurement. The updated system processed the IR readings during the robot's movement to detect when it passed over the shaded strip to determine its width and contrast. The robot was driven forward at a controlled speed so that sensor measurements could be linked reliably to position. Encoder feedback was used to estimate the robot's displacement, while the PID controller maintained a heading close to 0 radians to reduce lateral drift and improve measurement accuracy. This stage extends the system from offline data collection to active perception during robot operation.

The updated system shown in Algorithm 2 processes IR sensor readings in real time to detect when the robot passes over the shaded region. By identifying transitions between lighter and darker regions, the system determines both the presence of an anomaly and its spatial extent. To further improve robustness, a two-pass scanning strategy with backtracking was implemented. During the first pass the robot performs an initial detection of potential anomaly regions. After detection, the robot reverses to a position prior to the detected anomaly and performs a second pass using refined detection criteria. This approach improves consistency in identifying anomaly boundaries and reduces the impact of noise on detection accuracy.

C. Signal Processing

IR sensor readings are subject to noise and variability which can reduce the reliability of anomaly detection and boundary estimation. To address this, different signal

filtering techniques were investigated to improve detection performance. Three types of filter were selected and evaluated which were simple moving average (boxcar [3]), exponential moving average (EMA [4]) and a second-order Butterworth filter [5]. These filters represent increasing levels of complexity for comparison and were initially trialled against the baseline map.

III. Experiment Methodology

In this research project, experiments have been conducted on printed maps (shown in the GitHub Repository) with surface imperfections of prescribed width (1–5cm) and contrast (20–100% black). Two phases were conducted: an initial characterisation phase and a refinement phase focused on minimum detectable defect size. The experiments involve quantifying the robot's measurement error against the known imperfection characteristics.

A. Overview of Method

Testing Phase: Initially, simple maps were used to identify what the nominal readings from the IR sensors on the robot looked like when measuring the deepest contrasts achievable on printer paper. This then allowed for the calibration of the measurements to determine what the range of possible readings could be. From here, new maps were designed and printed such that they displayed zones of known width, with different levels of contrast % (10% to 20% contrast) in many different intervals. This latter stage was used to identify how the response of the sensors varied across the greyscale spectrum from the lightest value to the darkest value, and whether there was any difference in sensitivity between the two ends of the spectrum. The initial intervals used were of 10%, this was then dropped down to 5% and 3%. Tests were carried out at the extremities of the spectrum as well as in the centre to identify if the sensitivity peaked in the centre rather than at one of the ends. An example of a map can be seen in Figure 1, detailing the progression between stages of this part of the methodology.

Refinement Phase: In this phase of testing, the goal was shifted from assessing the capabilities of the sensors onboard the robot to extracting the maximum possible performance from them. To begin with, maps were designed with just one 'imperfection', being a bar of differing sizes and contrast values. The robot would perform two passes on this bar (forwards and backwards), but with a higher sampling rate to that used in the testing phase, as these samples were no longer stored on the robot but instead used to identify the crack size and contrast %. The initial contrast values used were 100% down to 40% in 20% intervals, and these were chosen initially to allow for a good differentiation between different 'imperfections'. Following the passes on the 'imperfection' the robot would then record the contrast and size, and store this value for analysis. The results of these tests were then used to refine the code used on the robot to improve its performance, with the final objective being to identify 'imperfections' with as minimal contrast value and size as possible.

B. Discussion of Variables

- **Independent Variable(s)**: Traversal speed was the primary independent variable, with the aim of improving the robot's ability to detect surface changes by increasing boundary sampling density. The contrast value of the imperfections was also varied independently. By determining the response of the IR sensors across the full contrast range, it was possible to understand how sensor sensitivity varied at different ends of the spectrum, allowing crack size analysis to be interpreted relative to the total capability of the sensor. Finally, imperfection size was varied as part of the refinement phase, to identify the smallest detectable crack achievable with the onboard sensors. Each variable was varied systematically while others were held constant, ensuring that observed changes in sensor response could be attributed to the variable under investigation rather than confounding factors.
- **Dependent Variable(s)**: The response of the robot's IR sensors to imperfection contrast value and size, represented by two features of the signal: the width of the response and its peak value. During the testing phase these were stored onboard and analysed after multiple trials to determine whether the robot could distinguish decreasing differences in contrast value. During the refinement phase, real-time processing replaced offline storage, and the primary focus shifted to determining the minimum imperfection reliably detectable by the sensors. This minimum detection threshold is the key dependent variable of interest, as it defines the practical operating limit of the system and determines the range of inspection scenarios to which this sensor method could be applied.
- **Controlled Variables**: Surface baseline reflectance was a critical controlled variable. The printer paper used was slightly translucent, causing readings that should have registered as zero for pure white to show non-negligible values. Extra white sheets were placed beneath the surface sheet to ensure the baseline read as close to zero as possible, ensuring consistency across all trials and allowing direct cross-trial comparison. Battery level was maintained by fully charging the robot prior to each data collection session, as voltage variation could affect motor speed and therefore sampling density. All experiments were conducted in the same room to ensure ambient IR conditions remained constant throughout, preventing environmental variation from introducing systematic error into the sensor baseline.

C. Discussion of Metric(s)

The initial testing phase was intentionally exploratory, with no quantitative metrics defined. This stage was designed to characterise the qualitative behaviour of the IR sensors, including signal responsiveness, noise characteristics, and sensitivity to changes in surface contrast. The primary objective was therefore to understand the signal characteristics prior to defining how performance would be formally evaluated.

For the refinement phase, two quantitative performance metrics were introduced to evaluate detection accuracy relative to the known properties of the map imperfections. Mean width error, taken across all five sensors and calculated as shown in Equation 1, measures spatial

detection accuracy proportionally to the true imperfection size, allowing direct comparison across different target widths.

$$WidthError(\%) = \frac{|W_{measured} - W_{true}|}{W_{true}} \times 100 \quad (1)$$

Additionally, mean peak value at each width was recorded to identify whether the robot correctly estimated imperfection contrast. These two metrics together capture both the spatial and photometric dimensions of detection performance. They were chosen because representing error as a proportion of the true value allows meaningful comparison across the full width and contrast test matrix, providing a complete characterisation of sensor capability and filter effectiveness across all tested conditions.

IV. Results

A. Baseline Sensor Characterisation (Algorithm 1)

Initial experiments were conducted using Algorithm 1 to collect raw IR sensor data as the robot traversed surfaces with predefined imperfections. This stage was intended for sensor characterisation rather than real-time detection.

The recorded sensor readings showed clear variations with surface reflectance, confirming the ability of the IR sensors to distinguish between background and defect regions. However, transitions between regions were gradual rather than discrete, indicating that the sensor integrates reflectance over a finite spatial footprint, producing a smoothing effect at boundaries.

The raw signal also exhibited noticeable noise, which limits the reliability of simple threshold-based detection. In addition, storing all readings onboard introduced memory constraints, limiting achievable sampling rate and scan duration. This highlights a trade-off between spatial resolution and data length.

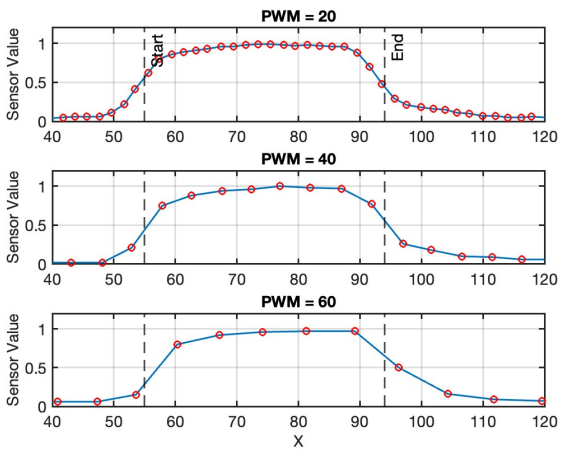

Fig. 3. Effect of robot traversal speed (PWM = 20, 40, 60) on IR sensor response and boundary sampling density.

B. Effect of Robot Speed on Sensor Response

The influence of robot traversal speed was evaluated using a fixed-width, 100% contrast target at different PWM settings. The results show that speed significantly affects spatial resolution. At higher PWM values, fewer samples are collected per unit distance, producing a sparse signal with reduced boundary detail. In contrast, lower PWM values increase sampling density and produce a more continuous signal. This is supported by boundary analysis, Fig. 3, where approximately 6 samples were recorded at PWM = 20, compared to 2 at PWM = 40 and 1 at PWM = 60. Reduced sampling density leads to poorer boundary reconstruction, as fewer points are available to approximate gradient transitions. These results confirm that slower motion improves detection accuracy by increasing spatial sampling density, enabling more reliable boundary localisation.

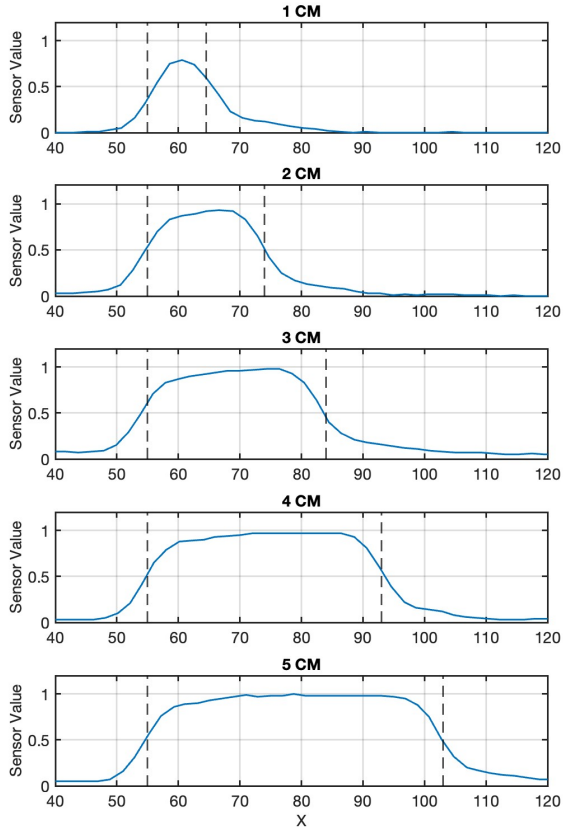

Fig. 4. Effect of imperfection size (5 cm, 4 cm, 3 cm, 2 cm, 1 cm) on IR sensor response showing spatial averaging and signal attenuation for smaller targets.

C. Effect of Imperfection Size on Sensor Response

The effect of imperfection size was evaluated using targets of varying widths at constant speed and 100% contrast. For larger targets (5 cm to 2 cm), the sensor response is stable and boundary transitions occur at approximately 50% of the calibrated signal. This reflects spatial averaging, where the sensor output corresponds to partial coverage of its field of view. This behaviour is clearly observed in Fig. 4, where larger targets produce consistent plateau regions and well-defined transitions.

For smaller targets (1 cm), the peak response is reduced and does not reach expected full-scale values. This is due to the sensor's finite spatial footprint, where both target and background are simultaneously measured, resulting in signal attenuation and blurred boundaries. These results

indicate a lower bound on detectable feature size imposed by sensor geometry rather than signal processing.

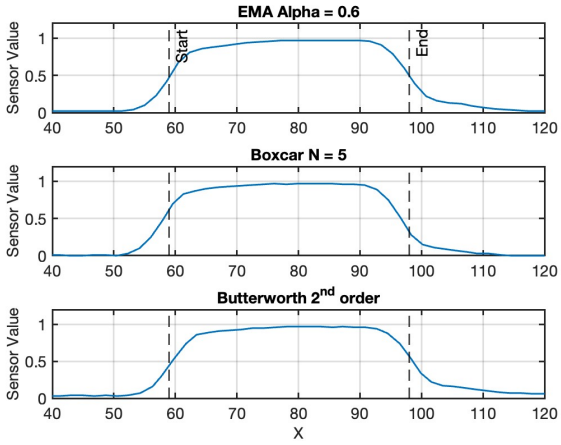

Fig. 5. Comparison of signal filtering methods: boxcar, exponential moving average (EMA), and Butterworth filter.

D. Effect of Signal Filtering on Sensor Response

Three filtering methods were evaluated against the baseline map: boxcar averaging, exponential moving average (EMA), and second-order Butterworth filtering. All three produced broadly similar overall signal shapes (Figure 5), confirming that the dominant factor in sensor behaviour is the physical IR response rather than electrical noise. This finding partially challenges the second part of the hypothesis, which anticipated filtering as an independent means of improving detection. The boxcar filter smoothed the signal but introduced boundary smearing and an effective phase lag, causing a spatial shift in detected feature positions. The Butterworth filter provided the strongest smoothing but at significant computational cost, therefore reducing loop execution rate and lowering effective spatial sampling density. This counteracted the benefit of slower traversal speed. The EMA filter with $\alpha = 0.6$ offered the best overall balance, weighting recent samples more heavily than past ones reduced noise without introducing significant phase lag and preserving boundary localisation while maintaining loop speed. These results reveal an important coupling between filtering choice and motion control where computational overhead from filtering can reduce the sampling rate benefit gained by slowing the robot. Thus, the two factors must be optimised jointly rather than independently.

E. Design Implications for Two-Pass IR Surface Anomaly Detection with Backtracking (Algorithm 2)

The results from Algorithm 1 directly informed the design of the two-pass detection strategy. The second-pass speed was set to the lowest stable PWM value, as slower motion increases sampling density and improves boundary resolution. The detection threshold was set to 50% of the peak value observed in the first pass, based on consistent boundary behaviour observed in larger targets. EMA filtering with smoothing factor $\alpha = 0.6$ was selected for the second pass due to its balance of responsiveness, noise reduction, and minimal phase lag compared to alternative filters. Overall, Algorithm 2 is directly parametrised using empirical relationships identified in Algorithm 1, ensuring improved boundary estimation through increased sampling density, adaptive threshold, and stable real-time filtering.

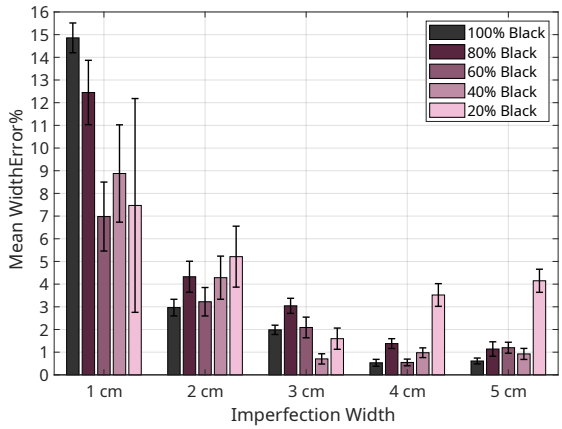

Fig. 6. Mean width error% varying imperfection width and contrast

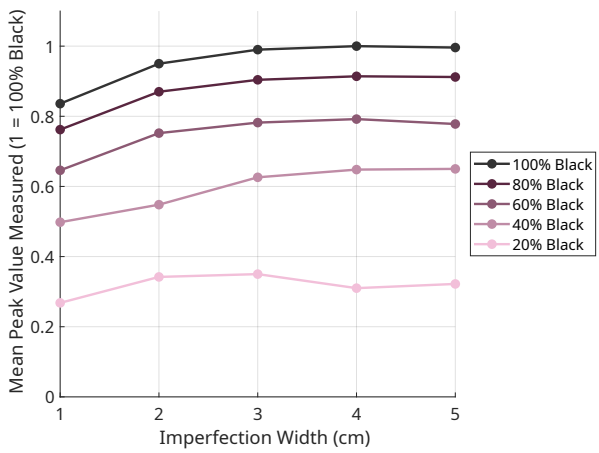

Fig. 7. Mean peak value measured varying contrast and imperfection width

F. Effect of Contrast and Imperfection Width on Detection Accuracy

Figures 6 and 7 present the combined effect of imperfection width and contrast on detection accuracy. Mean width error was highest at 10mm across all contrast levels, as the sensor footprint spans the entire feature and prevents reliable boundary detection. Error decreased progressively with increasing width, stabilising at widths of 40mm and above, indicating robust and consistent detection beyond this threshold. Contrast emerged as a secondary but significant factor. At 100% contrast, width error remained low across all detectable sizes, with clearly defined signal peaks approaching full-scale values. As contrast decreased to 20%, error increased substantially, particularly at smaller widths where the reduced signal amplitude approached the sensor noise floor, making threshold-based boundary detection unreliable. At intermediate contrast levels (40–80%), performance degraded gradually with decreasing width, suggesting a continuous rather than discrete relationship between contrast and detection limit. These results confirm that detection performance depends jointly on both size and

contrast; neither factor alone determines the detection limit, and any practical deployment of this system would need to account for both when specifying minimum detectable defect characteristics.

V. Discussion and Conclusion

A. Speed Reduction

The results from Section IV strongly support the first part of the hypothesis (Section I). Reducing traversal speed from $PWM = 60$ to $PWM = 20$ increased boundary samples from approximately 1 to 6, producing a measurably more reliable and continuous signal at anomaly boundaries. This improvement arises not from any change in sensor capability but from more effectively exploiting existing capability. Slower motion provides more sampling opportunities across the sensor's fixed spatial footprint, improving boundary reconstruction. The two-pass strategy in Algorithm 2 operationalises this insight by concentrating the low-speed scan specifically at detected regions of interest, avoiding the time cost of scanning the entire surface at reduced speed. This represents an efficient motion control solution that extracts near-maximum performance from a low-cost sensor without hardware modification.

B. Signal Filtering

The results show more ambiguity regarding the second part of the hypothesis. Although all three filters reduced high-frequency noise, they produced broadly similar signal shapes, suggesting that sensor behaviour is governed by the IR sensor's physical response rather than electrical noise. While filtering smoothed the signal, it could not recover spatial detail that was never captured, a fundamental constraint arising from the sensor's finite footprint geometry. The hypothesis therefore overestimated the role of filtering in improving detection performance; noise reduction alone is necessary but not sufficient for reliable anomaly boundary detection. The choice of filtering strategy did have important consequences. The second-order Butterworth filter provided the strongest smoothing (Figure 5) but introduced greater computational overhead, reducing loop execution rate and effectively lowering spatial sampling, critical for boundary detection. This revealed an overlooked coupling: reducing traversal speed only improves sampling density if the control loop can run fast enough to use the extra time. The EMA filter ($\alpha = 0.6$) offered the best balance of noise reduction and boundary preservation without the phase lag of the boxcar filter or the overhead of the Butterworth filter. A refined hypothesis would treat speed and filtering as interdependent rather than separate improvements.

C. Limitations

The most significant limitation identified during experimentation was the sensitivity of baseline calibration to subsurface reflectance. The printer paper used was discovered to be slightly translucent, causing sensor readings to be influenced by the surface beneath the map. This produced an approximately 10% contrast offset visible in Figure 7, where measured contrast consistently exceeded the true value. Whilst this was rectified experimentally by placing additional white sheets beneath the map, this solution would not generalise to real-world surface inspection applications where the reflectance of the underlying environment cannot be controlled or standardised. In such scenarios, the baseline calibration procedure would need to be adapted dynamically, potentially using an initial calibration pass over a known reference region before commencing inspection. This represents a non-trivial engineering challenge that would need to be addressed before this system could be deployed practically.

A further limitation concerns the uniformity of the test environment. All experiments were conducted on flat, printed maps with precisely defined imperfection characteristics. Real surface defects such as cracks or discontinuities are rarely uniform in width, contrast, or geometry, and may occur on curved, textured, or otherwise non-uniform surfaces. The extent to which detection performance degrades under these conditions remains uncharacterised. Additionally, the robot's lateral drift, managed in this study through PID heading control, may be more problematic on non-uniform surfaces where the heading reference itself may be unreliable. These constraints represent a significant gap between laboratory results and practical applicability that future work should address.

D. Conclusion

This research investigated whether a two-pass scanning strategy combined with signal filtering could improve surface anomaly detection using low-cost IR reflectance sensors. Reliable detection was achieved down to 2cm at 100% contrast, with performance degrading predictably at lower contrast and smaller feature sizes and consistent with the sensor's finite spatial integration area. The two-pass approach validated the core hypothesis, with EMA filtering ($\alpha = 0.6$) providing the best balance of noise reduction and loop speed. However, filtering alone could not recover spatial detail never captured, confirming speed and filtering as interdependent improvements. The primary limitation lies with sensor geometry; features smaller than 1–2cm require a more collimated sensing modality. Within the sensor's operating range however, the two-pass strategy demonstrates that meaningful performance gains are achievable through motion control alone, potentially negating the need for more expensive hardware.

Future work should focus on directly characterising the IR sensors' spatial footprint so detection limits can be predicted analytically, replacing the empirical approach used in Algorithm 1. For oblique or narrow features, combining all five sensors could improve detection. Further testing on non-uniform surfaces would assess calibration robustness and show how performance degrades when baseline reflectance varies, providing a more realistic evaluation.

References

[1] D. Grabowski and C. Cristalli, "Production line quality control using infrared imaging," *Infrared Physics & Technology*, vol. 71, pp. 416–423, 2015.

[2] A. Alhaddad, A. Al-Araji, and Z. Al-Qassar, "Investigation of infrared sensors for robot navigation," *International Journal of Engineering Research and Technology*, vol. 12, no. 9, pp. 1506–1512, 2019.

[3] S. W. Smith, *The Scientist and Engineer's Guide to Digital Signal Processing*. San Diego, CA: California Technical Publishing, 1997.

[4] C. C. Holt, *Forecasting Seasonals and Trends by Exponentially Weighted Moving Averages*. Pittsburgh, PA: Carnegie Institute of Technology, 1957.

[5] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms and Applications*. New Jersey: Pearson Prentice Hall, 4th ed., 2007.